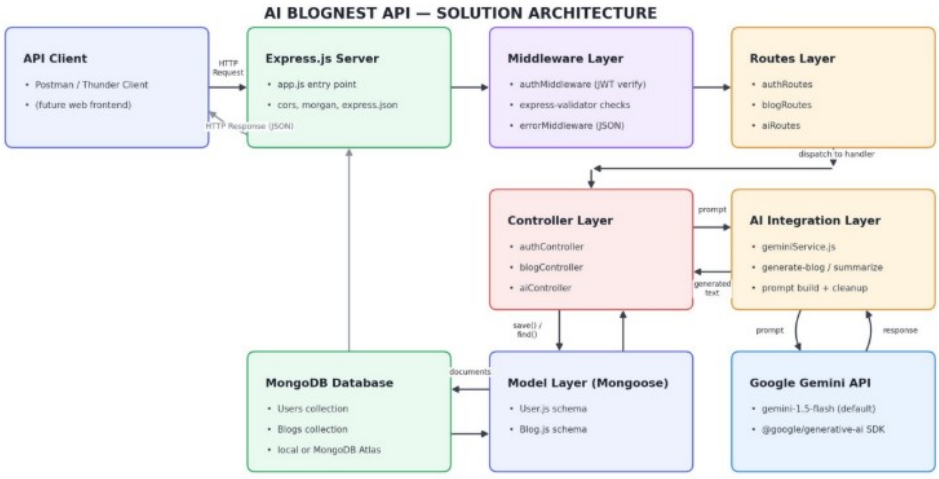
Project Design Phase
Solution Architecture

Date	28 September 2026
Team ID	SWTID-2026-2598
Project Name	AI BlogNest API
Maximum Marks	4 Marks

Solution Architecture Diagram – AI BlogNest API



API clients (Postman / a future frontend) send HTTP requests to the Express.js server. Requests pass through a middleware layer that handles JWT authentication, role checks, request validation, and error handling, before reaching the routes layer (Auth, Blog, and AI routes).

The controller layer processes business logic and talks to two downstream systems: the Model layer (Mongoose schemas for Users and Blogs, persisted in MongoDB) and the AI Integration layer, which calls Google's Gemini API to generate blog drafts from a topic or summarize existing content. Generated content flows back through the controller layer and is persisted to MongoDB in the same way as manually written content.